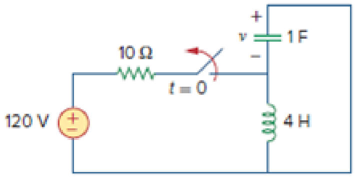
Chapter 8, Problem 19P

Problem

Obtain $v(t)$ for $t > 0$ in the circuit of Fig. 8.73.

Figure 8.73:



Step-by-step solution

Step 1 of 6

At t < 0; we have,

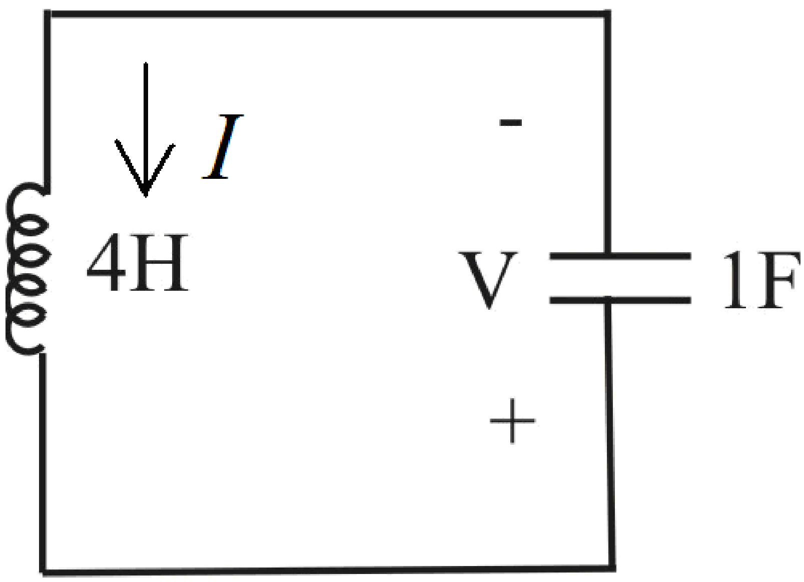
$$i_L(0^-) = \frac{120}{10}$$

$$i_L(0^-) = 12\,\text{A}$$

$$V_C(0^-) = 0\,\text{V}$$

Step 2 of 6

At t > 0; we have circuit,



Step 3 of 6

Since R = 0, $\alpha = 0$,

Therefore $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\omega_0 = \frac{1}{\sqrt{4 \times 1}}$$

$$\omega_0 = \frac{1}{2}$$

$$\omega_0 = 0.5\,\text{rad/s}$$

Therefore $\omega_d = 0.5\,\text{rad/s}$

Step 4 of 6

The solution is of the form

$$v(t) = e^{-\alpha t}\left(B_1 \cos\omega_d t + B_2 \sin\omega_d t\right)$$

Since $\alpha = 0$ and $\omega_d = 0.5$

$$v(t) = B_1 \cos 0.5t + B_2 \sin 0.5t$$

At t = 0 ,

$$v(0) = B_1$$

$$B_1 = 0$$

Therefore $v(t) = B_2 \sin 0.5t$

Step 5 of 6

Now $\frac{dv(t)}{dt} = 0.5B_2 \cos 0.5t$

The current through the capacitor is

$$i = C\frac{dv}{dt}$$

$$\frac{dv}{dt} = \frac{i}{C}$$

At $t = 0^+$, the current through the inductor is flowing through the capacitor is

$$\frac{dv(0^+)}{dt} = \frac{i(0^+)}{C}$$

$$\frac{dv(0^+)}{dt} = \frac{12}{1}$$

$$\frac{dv(0^+)}{dt} = 12$$

Step 6 of 6

At $t = 0^+$

$$\frac{dv(0^+)}{dt} = 0.5B_2$$

$$0.5B_2 = 12$$

$$B_2 = 24$$

Therefore the voltage across the capacitor is,

$$\boxed{v(t) = 24\sin 0.5t \text{ V}}$$